Table 7.1 The expected long-time computed average using the given second-order Langevin dynamics discretization scheme

Scheme	$\langle q^2\rangle_h/\langle q^2\rangle$	$\langle p^2\rangle_h/\langle p^2\rangle$	$\langle qp\rangle$
⟦ABOBA⟧	1	$[1-h^2\Omega^2/(4m)]^{-1}$	0
⟦OBABO⟧	$[1-h^2\Omega^2/(4m)]^{-1}$	1	0
⟦BAOAB⟧	1	$1-h^2\Omega^2/(4m)$	0
⟦OABAO⟧	$1-h^2\Omega^2/(4m)$	1	0

schemes, we are able to compute the exact expected average by solving the linear equations. The results are summarized in Table 7.1, where we have not considered the ⟦AOBOA⟧ and ⟦BOAOB⟧ schemes due to their inconsistency at large friction (this does not preclude the analysis, but limits each those method's usefulness in simulation). Similar analysis can be conducted for more complicated composition schemes involving multiple force evaluations.

The mentioned splitting schemes give a second-order error in the computed averages, but surprisingly each gives an exact result for computed averages of either position or momentum. The exactness property is not born out of the symmetry of the method, but rather of the specific order of terms and choice of splitting "building blocks". Even non-symmetric first-order schemes can exhibit similar qualities. For example, analysis shows that the schemes denoted ⟦AOAB⟧ or ⟦BAOA⟧ both exhibit a first order error in the covariance $\langle qp\rangle_h = \mathcal{O}(h)$, however they give exact results for both $\langle q^2\rangle_h$ and $\langle p^2\rangle_h$.

7.4.2 More General Schemes

We can conduct a similar analysis for any integrator that we can express in the form (7.7), simply by plugging in the relevant constants and solving a linear equation. The difficulty lies when random numbers are reused, leading to correlation between steps. This requires careful treatment to compute the expectation.

Example 7.2 Consider the variant of the BBK integrator that recycles the random number between steps:

Brünger-Brooks-Karplus with post-step re-use of random number (BBK*)

$$\boldsymbol{p}_{n+1/2} = (1-h\gamma/2)\boldsymbol{p}_n - (h/2)\nabla U(\boldsymbol{q}_n) + \tfrac{1}{2}\sqrt{2k_{\mathrm{B}}\mathrm{T}h\gamma}\boldsymbol{M}^{1/2}\mathbf{R}_n$$
$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2}$$
$$\boldsymbol{p}_{n+1} = [\boldsymbol{p}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_{n+1}) + \tfrac{1}{2}\sqrt{2k_{\mathrm{B}}\mathrm{T}h\gamma}\boldsymbol{M}^{1/2}\mathbf{R}_{n+1}]/(1+h\gamma/2),$$

where $\mathbf{R}_n$ is a vector of normal random numbers with zero mean and unit variance. In some systems, it can be computationally expensive to generate normal random numbers for each degree of freedom, hence algorithms may re-use the random

numbers between iterations to give a small computational saving. We can see that only one vector of random numbers is generated per iteration, with the new vector of random numbers being used in the following iteration. This gives some correlation between steps that complicates our analysis a little.

If we wish to find the perfect sampling bias in the case of the one-dimensional harmonic oscillator, we can apply this technique to find the invariant averages under this scheme. We have

$$\Psi^{\mathrm{BBK}^*} = \begin{bmatrix} 1 - \frac{h^2\Omega^2}{2m} & \frac{h}{m}\left(1 - \frac{\gamma h}{2}\right) \\ \frac{h\Omega^2}{2m}\left(\frac{h^2\Omega^2 - 4m}{2+\gamma h}\right) & \frac{(2-\gamma h)\left(2m - h^2\Omega^2\right)}{2m(2+\gamma h)} \end{bmatrix},$$

$$\mu_n^{\mathrm{BBK}^*} = \begin{bmatrix} \sqrt{\frac{k_\mathrm{B}\mathrm{T}\gamma h^3}{2m}}\mathrm{R}_n \\ \frac{\sqrt{2k_\mathrm{B}\mathrm{T}\gamma h m}}{2+\gamma h}\left(\left(1 - \frac{h^2\Omega^2}{2m}\right)\mathrm{R}_n + \mathrm{R}_{n+1}\right) \end{bmatrix} = \begin{bmatrix} \kappa_1 \mathrm{R}_n \\ \kappa_2 \mathrm{R}_n + \kappa_3 \mathrm{R}_{n+1} \end{bmatrix},$$

using the notation in (7.7). Taking expectations of the resulting equations, we can compute the values of the constants in (7.8)–(7.10) and solve to find the long-time averages. Care must be taken however, as unlike the schemes using the ABO splitting we must include some correlation between steps. Denoting the entries of Ψ^{BBK^*} by ψ_{ij} we have

$$\begin{aligned} \mathbb{E}(\mu_{n,2} p_n) &= \mathbb{E}\left(\left(\kappa_2 \mathrm{R}_n + \kappa_3 \mathrm{R}_{n+1}\right)\left(\psi_{21} q_{n-1} + \psi_{22} p_{n-1} + \kappa_2 \mathrm{R}_{n-1} + \kappa_3 \mathrm{R}_n\right)\right), \\ &= \kappa_2 \kappa_3. \end{aligned}$$

This additional quantity will slightly change the expectation of (7.9) compared to the standard BBK method, and hence the overall averages of the scheme. Additionally it is easily shown that

$$\mathbb{E}(\mu_{n,1} q_n) = \mathbb{E}(\mu_{n,2} q_n) = 0, \quad \mathbb{E}(\mu_{n,1} p_n) = \kappa_1 \kappa_3,$$

with

$$\mathbb{E}(\mu_{n,1}^2) = \kappa_1^2, \quad \mathbb{E}(\mu_{n,1}\mu_{n,2}) = \kappa_1 \kappa_2, \quad \mathbb{E}(\mu_{n,2}^2) = \kappa_2^2 + \kappa_3^2.$$

Thus taking expectations of (7.8)–(7.10) and substituting in the constants gives

$$\begin{aligned} \langle q^2 \rangle_h^{\mathrm{BBK}^*} &= \psi_{11}^2 \langle q^2 \rangle_h^{\mathrm{BBK}^*} + \psi_{12}^2 \langle p^2 \rangle_h^{\mathrm{BBK}^*} + 2\psi_{11}\psi_{12} \langle qp \rangle_h^{\mathrm{BBK}^*} + 2\psi_{12}\kappa_1 2\psi_{12}\kappa_1\kappa_3 \\ &\quad + \kappa_1^2, \\ \langle p^2 \rangle_h^{\mathrm{BBK}^*} &= \psi_{21}^2 \langle q^2 \rangle_h^{\mathrm{BBK}^*} + \psi_{22}^2 \langle p^2 \rangle_h^{\mathrm{BBK}^*} + 2\psi_{21}\psi_{22} \langle qp \rangle_h^{\mathrm{BBK}^*} + 2\psi_{22}\kappa_2 2\psi_{22}\kappa_2\kappa_3 \\ &\quad + \kappa_2^2 + \kappa_3^2, \end{aligned}$$